

André Rosenbaum Coelho - CV

Orcid: <https://orcid.org/0009-0003-8738-0770>

Webpage: <https://andrerosenbaumcoelho.github.io/>

Contact: andre.rosenbaum2002@gmail.com/ andrerosenbaumcoelho@usp.br

1. Education

Period	Program	Institution	Advisor(s)
2017–2019	High School	Colégio Santa Cruz (São Paulo, Brazil)	—
2020–2024	B.Sc. in Mathematics Education	IME-USP (Institute of Mathematics and Statistics, University of São Paulo, Brazil)	Prof. Sinai Robins
2025-2027	Master's in Pure Mathematics	IME-USP/Université de Montréal (Montréal, Canada)	Prof. Sinai Robins Supervision in Montréal: Prof. Andrew Granville

2. Professional and Academic Experience

1. Teaching Assistant for **Differential and Integral Calculus I (MAT2453)**, Institute of Mathematics and Statistics, University of São Paulo. Workload: 6 hours/week. (March-July 2026)
2. Teaching Assistant for **Elementary Number Theory (MAT0164)**, Institute of Mathematics and Statistics, University of São Paulo. Workload: 6 hours/week. (Aug-Dec 2025)
3. Teaching Assistant for **Linear Algebra I (MAT0122)**, Institute of Mathematics and Statistics, University of São Paulo. Workload: 6 hours/week. (Aug-Dec 2023)

4. Mathematics teacher at **Cursinho Popular Emancipa Queixadas**. Bimonthly (May 2019-December 2024)

3. Research

Publications

1. Valle, J. C., Maciel, I. B., Tersariolli, P. M., Coelho, A. R. *Mathematics Education in the Context of Quilombola School Education in Two Bibliographical Surveys*. **TANGRAM - Revista de Educação Matemática**, 6(3), 167–189, 2023. DOI: [10.30612/tangram.v6i3.17530](https://doi.org/10.30612/tangram.v6i3.17530)

Preprints

1. Coelho, A., Robins, S. Half-open integer parallelepipeds and polytope Dedekind sums (2026), submitted. DOI <https://doi.org/10.48550/arXiv.2608.18408>
2. Coelho, A., Robins, S., Simon, C. *Rational Eigenfunctions of the Hecke Operators* (2024), submitted. DOI [10.48550/arXiv.2406.15744](https://doi.org/10.48550/arXiv.2406.15744)

4. Grants and Prizes

Grants:

1. **FAPESP Master's Scholarship**
 - Monthly funding: R\$3,270.00
 - Grant number: 2025/06117-4
 - Period: Nov. 1, 2025 – Feb. 28, 2027
 - Associated with the thematic project *Classical, Asymptotic, Quantum, and Geometric Combinatorics* (2023/03167-5)
 - BEPE: Grant for research internship abroad: R\$16.440 + USD 8.598
 - 4 months at the Université de Montréal. Supervision: Prof. Andrew Granville
2. **CAPES-PROEX Master's Scholarship**
 - Monthly funding: R\$2,100.00
 - Grant number: 88887.141114/2025-00

- Period: Mar. 1, 2025 – Oct. 31, 2025
3. **PAE Scholarship — Teaching Assistant for MAT2453 (Differential and Integral Calculus I)**
 - Monthly funding: R\$1,000.00
 - Period: Feb. 1, 2026 – Jun. 30, 2026
 4. **PAE Scholarship — Teaching Assistant for MAT0164 (Elementary Number Theory)**
 - Monthly funding: R\$960.00
 - Period: Jul. 1, 2025 – Nov. 30, 2025
 5. **ICMC-USP Research Program Scholarship (REU)**
 - **Project:** *Investigating Lattice Points and Volume Relationships in 3-Dimensional Simplexes*
 - **Supervisors:** Prof. Ali Tahzibi (ICMC-USP) and Prof. Javid Validashti (DePaul University)
 - Funding: R\$2,000.00
 - Period: Jan. 10, 2025 – Feb. 18, 2025
 6. **IMPA Summer Scholarship**
 - Funding: R\$2,400.00
 - Period: Jan. 2, 2024 – Mar. 1, 2024
 7. **Unified Scholarship Program (PUB)**
 - **Project:** *Mathematics in Quilombola School Curricula and Curriculum Development in Barra do Turvo, São Paulo.*
 - **Supervisor:** Prof. Júlio César Augusto do Valle
 - Monthly funding: R\$500.00
 - Period: Apr. 27, 2023 – Aug. 31, 2023

Prizes:

1. **“Prêmio Hildebrando Munhoz Rodrigues de Iniciação Científica da SBM, Nível prata, categoria Apresentação Oral” (ICMC-USP, 2024)**, for the second best oral presentation on undergraduate research: “A ação de um operador de Hecke no espaço das funções racionais”. R\$1,000.00. [Link for the presentation.](#)

6. Additional Information

Research Programs

1. **ICMC-USP Research Program for Undergraduates/Early Master's students (Summer 2025)**
 - **Project:** *Investigating Lattice Points and Volume Relationships in 3-Dimensional Simplexes*
 - **Supervisors:** Prof. Ali Tahzibi (ICMC-USP) and Prof. Javid Validashti (DePaul University)
 - During this program, research was conducted on open problems in mathematics. In particular, developing lower bounds for the number of lattice points in half-open tetrahedra in terms of its volume.

Summer Courses

1. **Introduction to Modular Forms and Elliptic Curves (IMPA, 2026)**
 - **Instructor:** Prof. Hossein Movasati
2. **Introduction to Number Theory (IMPA, 2024)**
 - **Instructor:** Prof. Carlos Gustavo Moreira (Gugu)

Participation in Academic Events

1. **Québec-Maine Number Theory Conference** (October 2026)
ICMC-USP, São Carlos, Brazil
Poster presentation: "Manifestations of Barnes Polynomials in Lattice Point Enumeration in Polytopes"
2. **XXVIII Brazilian Algebra Meeting** (July 2026)
ICMC-USP, São Carlos, Brazil
Poster presentation: "Manifestations of Barnes Polynomials in Lattice Point Enumeration in Polytopes"
3. **Pure Mathematics Week** (May 2026)
IME-USP, São Paulo, Brazil
Poster presentation: "Manifestations of Barnes Polynomials in Lattice Point Enumeration in Polytopes"

4. **32nd SIICUSP** (October 2024)
IME-USP, São Paulo, Brazil
Poster presentation: "Rational Eigenfunctions of the Hecke Operators"

5. **4th Brazilian Mathematics Colloquium of the Southeast Region** (October 2024)
IBILCE-UNESP, São José do Rio Preto, Brazil
Oral presentation in the thematic session "Algebra": "Rational Eigenfunctions of the Hecke Operators"

6. **National Undergraduate Research Week** (August 2024)
ICMC-USP, São Carlos, Brazil
Oral presentation. Prize: Prêmio Hildebrando Munhoz Rodrigues de Iniciação Científica da Sociedade Brasileira de Matemática